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Abstract. Information security has attracted much current attention due to the rapid development of modern technologies, such as computer and internet. We propose a novel method for optical image encryption using multilevel Arnold transform and rotatable-phase-mask noninterferometric imaging. An optical image encryption scheme is developed in the gyrator transform domain, and one phase-only mask (i.e., phase grating) is rotated and updated during image encryption. For the decryption, an iterative retrieval algorithm is proposed to extract high-quality plaintexts. Conventional encoding methods (such as digital holography) have been proven vulnerable to the attacks, and the proposed optical encoding scheme can effectively eliminate security deficiency and significantly enhance cryptosystem security. The proposed strategy based on the rotatable phase-only mask can provide a new alternative for data/image encryption in the noninterferometric imaging. © 2011 Society of Photo-Optical Instrumentation Engineers (SPIE). [DOI: 10.1117/1.3643724]

Subject terms: optical image encryption; optical cryptosystem; noninterferometric imaging; rotatable phase mask; Arnold transform; security analysis.

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1 Introduction

The cryptography concept is related to the science of secret writing, and a cryptosystem is usually designed based on data conversion between plaintexts and ciphertexts. When one receives ciphertexts, he/she needs to apply correct security keys in order to successfully extract the hidden plaintexts.¹ Diffie and Hellman² divided the cryptosystems into two types, i.e., symmetric and asymmetric. In the symmetric cryptosystem, the same security keys are used for data encryption and decryption. Hence, the transmission or communication highly depends on the reliability of senders and receivers. In an asymmetric cryptosystem,² there exist two different keys, i.e., enciphering and deciphering keys.

In recent years, the optical method^{3,4} has emerged as an effective and powerful tool for image encryption due to its multidimensional and multiparameter capabilities. Since double random phase encoding was proposed,³ optical encryption has been widely studied for image hiding. Some algorithms and infrastructures,^{5–20} such as fractional Fourier transform^{5,6,9} and computational ghost imaging,⁷ have been further proposed to enhance cryptosystem security. The algorithms (such as Fresnel transform and fractional Fourier transform) can be easily implemented, while the algorithms, such as computational ghost imaging, are relatively complicated. However, due to the linearity property, some cryptosystems (such as Fresnel transform⁸) have been proven to vulnerably attack algorithms,^{21–24} such as known-plaintext attack.^{21,23} Recently, noninterferometric imaging^{25–27} has attracted much attention in the phase retrieval field due to its significant advantages, such as simple optical setup. The noninterferometric method is developed based on a single-path wave propagation strategy, and the extraction procedure can be stated as an inverse problem. In this paper, we propose a

novel method for optical image encryption using multilevel Arnold transform (ART) and rotatable-phase-mask noninterferometric imaging. An optical encryption scheme is developed in the gyrator transform domain, and one phase-only mask (i.e., phase grating) is rotated and updated during image encryption. For the decryption, an iterative retrieval algorithm is proposed to extract high-quality plaintexts. Conventional encoding methods (such as digital holography) have been proven to be vulnerable to the attacks, and the proposed optical encoding scheme can effectively eliminate security deficiency and significantly enhance cryptosystem security. The proposed strategy based on a rotatable phase-only mask can provide a new alternative for data/image encryption in the noninterferometric imaging.

2 Theoretical Analysis

Figure 1 shows a schematic setup for optical image encryption in the gyrator transform domain. A collimated plane wave illuminates an input image, and two independently random phase-only masks, M1 and M2, are placed in the optical path. For simplicity, only two phase-only masks are used, since it is straightforward to apply more phase-only masks. In addition, a phase grating G is placed just before the plaintext. Let $\exp[i\varphi_1(x', y')]$ and $\exp[i\varphi_2(\mu, \nu)]$, respectively, denote phase-only masks M1 and M2, where $i = \sqrt{-1}$, and $\varphi_1(x', y')$ and $\varphi_2(\mu, \nu)$ are two-dimensional (2D) maps randomly distributed in a range of $[0, 2\pi]$.

Before image encryption, a pixel scrambling algorithm, i.e., ART,^{28–30} is first applied to protect plaintext information. A discrete ART of the plaintext $O(x, y)$ is described by

$$\begin{aligned} \text{ART}[O(x, y), N] &= \{[v, (x', y')](x', y')^T \\ &= W(x, y)^T \pmod{N}, \\ [v, (x, y)] &\in O(x, y)\}, \end{aligned} \quad (1)$$

where $W = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, mod denotes the modulus after division, (x', y') is a new pixel position after ART operation, T denotes transpose operation, N is pixel number, v denotes a value at the pixel (x, y) in the original 2D map, and the terms on the right-hand side of “|” denote operation procedures.^{28–30} Since there is a scrambling period in the ART, decryption difficulty does not increase for authorized receivers.

In this study, multiple-level ART strategy is proposed to generate a scrambled 2D map, which may further enhance cryptosystem security. When the proposed multiple-level ART strategy is unknown during image decryption, it is impossible to extract the plaintexts. The multiple-level ART strategy proceeds as follows: 1. the whole plaintext information $O(x, y)$ (i.e., 512×512 pixels) is first scrambled by the ART; 2. the scrambled map obtained in step 1. is equally divided into four regions (i.e., each region of 256×256 pixels), and different number of iterations in the ART is set for each region; 3. the whole scrambled 2D map obtained in step

2. (i.e., 512×512 pixels) is further processed by the ART. The multiple-level ART strategy aforementioned is simple, and a design of more complicated multiple-level ART methods is straightforward. Subsequently, the scrambled plaintext $O(x', y')$ is further processed by the following optical encryption scheme. A schematic flow chart of multiple-level ART strategy is shown in Fig. 2.

Hence, ciphertexts $I^p(\xi, \eta)$ in the charge-coupled device (CCD) plane can be obtained by

$$I^p(\xi, \eta) = |GY_\beta\{GY_\alpha\{\exp[i g^p(x', y')]O(x', y')\} \times \exp[i \varphi_1(x', y')]\} \exp[i \varphi_2(\mu, \nu)]|^2, \quad (2)$$

where GY_α and GY_β , respectively, denote gyrator transforms with angles of α and β , integer p denotes a recording sequence (i.e., 1, 2, 3, ...), $| \cdot |$ denotes a modulus operation, and $g^p(x', y')$ denotes the rotated phase grating $G^p(x', y')$. Gyrator transform with an angle of α in Eq. (2) can be described by³¹

$$GY_\alpha\{\dots\} = \frac{1}{|\sin \alpha|} \iint \exp[i g^p(x', y')]O(x', y') \exp[i \varphi_1(x', y')] \times \exp\left[i 2\pi \frac{(x'y' + \mu\nu) \cos \alpha - (x'v + y'\mu)}{\sin \alpha}\right] dx' dy'. \quad (3)$$

Similarly, gyrator transform GY_β can also be described by Eq. (3). When angle α is equivalent to $\pi/2$, gyrator transform can be considered as Fourier transform. Gyrator transform possesses periodic and additive characteristics, i.e., $GY_\alpha = GY_{2\pi+\alpha}$ and $GY_{\alpha_1}GY_{\alpha_2} = GY_{\alpha_1+\alpha_2}$.

For image decryption, an iterative retrieval algorithm is developed and three ciphertexts (p amounts to 3) are applied. The algorithm consists of the following steps:

1. An initial random or constant map $\mathfrak{S}_n(x', y')$ (the number of iterations $n = 1, 2, 3, \dots$) is first assumed for the plaintext;
2. Multiply by phase grating and phase-only mask:

$$\mathfrak{S}'_n(x', y') = \mathfrak{S}_n(x', y') G^1(x', y') \exp[i \varphi_1(x', y')];$$

3. Propagate forward to the CCD plane:

$$O_n^1(\xi, \eta) = GY_\beta\{GY_\alpha[\mathfrak{S}'_n(x', y')] \exp[i \varphi_2(\mu, \nu)]\};$$

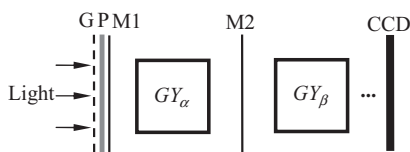


Fig. 1 A schematic setup for optical encryption in the gyrator transform domain: M1 and M2, phase-only masks; G, phase grating; P, plaintext; GY, gyrator transform; CCD, charge-coupled device.

4. Apply a modulus constraint using square root of the ciphertext:

$$\overline{O_n^1(\xi, \eta)} = [I^1(\xi, \eta)]^{1/2} [O_n^1(\xi, \eta) / |O_n^1(\xi, \eta)|];$$

5. Propagate back to the plaintext plane:

$$\mathfrak{S}_n^1(x', y') = [GY_{-\alpha}\{GY_{-\beta}[\overline{O_n^1(\xi, \eta)}] \times \{\exp[i \varphi_2(\mu, \nu)]\}^* \} \times \{\exp[i \varphi_1(x', y')]\}^* [G^1(x', y')]^*],$$

where the asterisk denotes complex conjugate;

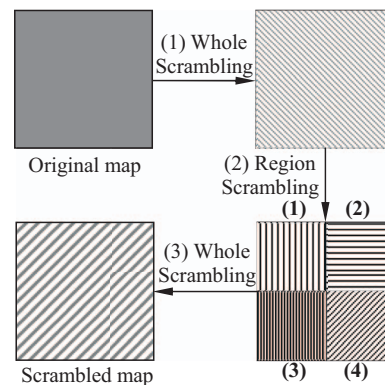


Fig. 2 Schematic flow chart for the proposed multiple-level ART strategy.

6. Repeat steps 2–5 using other parameters [i.e., $G^2(x', y')$ and $I^2(\xi, \eta)$; $G^3(x', y')$ and $I^3(\xi, \eta)$], and subsequently obtain $\mathfrak{I}_n^3(x', y')$;
7. Calculate an iterative error³² by using $\overline{\mathfrak{I}_n^3(x', y')}$ to decide whether iterative operation should be stopped. If the error is not larger than a preset threshold (δ), the iteration operation stops. Otherwise, complex amplitude $\overline{\mathfrak{I}_n^3(x', y')}$ is further used for the next iteration ($n = n + 1$) starting at step 2;
8. After the above iterative operation meets threshold requirement, a decrypted image can be obtained by $\overline{O(x, y)} = \text{MIART}[\overline{\mathfrak{I}_n^3(x', y')}]$, where MIART denotes multiple-level inverse Arnold transform. Since only an amplitude map is required, phase-only mask M1 can be neglected during the decryption.

Correlation coefficients (CC)^{29,32–34} can quantify the strength of a relationship between two variables (or matrixes), and are calculated to evaluate the similarity between the plaintext $O(x, y)$ and decrypted image $\overline{O(x, y)}$.

$$\text{CC} = \frac{\text{CRC}(O, \overline{O})}{\sigma_O \times \sigma_{\overline{O}}}, \quad (4)$$

where CRC denotes cross-covariance and σ denotes standard deviation.

3 Results and Discussion

A numerical experiment shown in Fig. 1 is conducted to demonstrate feasibility and effectiveness of the proposed method. The pixel size of CCD camera is $10 \mu\text{m}$, and pixel number is 512×512 . A collimated plane wave (light wavelength of 630 nm) is generated to illuminate a plaintext, and two independently random phase-only masks are placed in the optical path. The phase-only masks M1 and M2 are 2D maps randomly distributed in a range of $[0, 2\pi]$. Iteration number of 30 and 80 is, respectively, set at the first and third-step whole scrambling, and iteration number of 40, 50, 60, and 70 is set at region scrambling as shown in Fig. 2. The angles of α and β in the gyrator transform are set as $\pi/6$ and

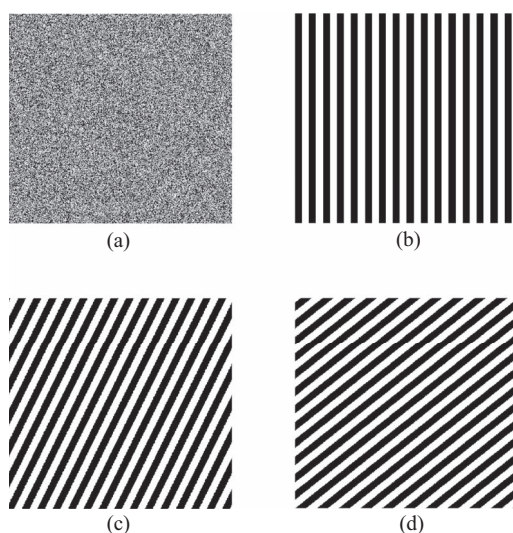


Fig. 3 (a) Phase-only mask M1, and (b)–(d) phase gratings $g^p(x', y')$ ($p = 1, 2, 3$).

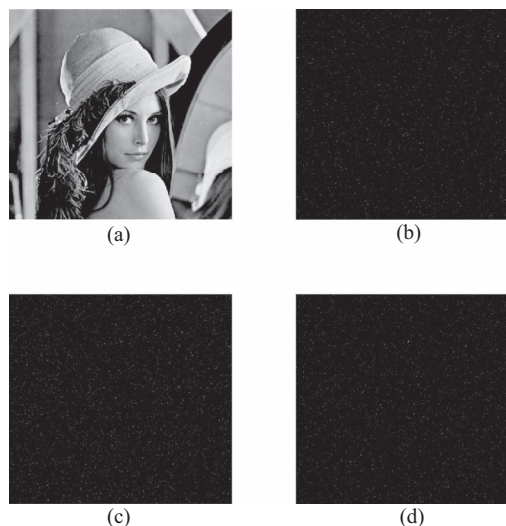


Fig. 4 (a) A plaintext, and (b)–(d) ciphertexts obtained when phase gratings in Figs. 3(b)–3(d) are used.

$\pi/3$, respectively. A threshold δ of 0.0001 is preset in the proposed retrieval algorithm during image decryption, and in some cases (such as occlusions) a higher threshold can be set.

Figure 3(a) shows a typical phase-only mask (i.e., M1 with 512×512 pixels), and Figs. 3(b)–3(d) show phase gratings $g^p(x', y')$ ($p = 1, 2, 3$). The grating is binary (i.e., 0 and $\pi/3$), and grating pitch is set as 32 pixels. Rotation of 25 and 50 deg are applied. Figure 4(a) shows a plaintext [Lena (Ref. 35) with 512×512 pixels], and Figs. 4(b)–4(d) show ciphertexts obtained by using phase gratings $G^p(x', y')$ ($p = 1, 2, 3$) during the encryption, respectively. It can be seen in Figs. 4(b)–4(d) that the plaintext is fully hidden, and no information about the input image can be observed. Figure 5(a) shows a relationship between the number of iterations and iterative errors (after natural logarithm), when correct security keys are used during image

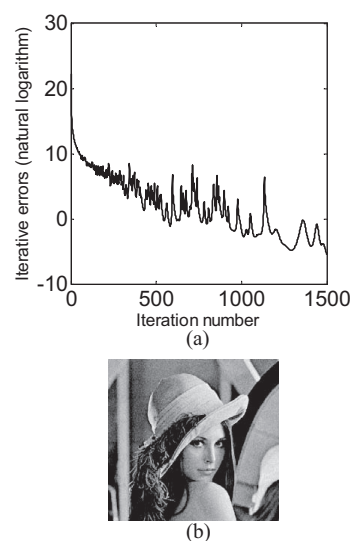


Fig. 5 Correct security keys: (a) a relationship between the number of iterations and iterative errors (after natural logarithm); and (b) a decrypted image.

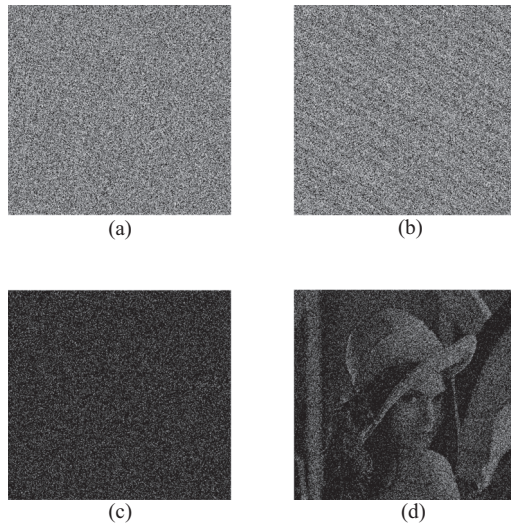


Fig. 6 Decrypted images (a) without the MIART, (b) using wrong iteration number in the MIART, and (c) using a wrong angle β ; (d) a decrypted image when all ciphertexts are occluded.

decryption. It can be seen in Fig. 5(a) that iterative errors can rapidly decrease to a satisfactory point. After 1500 iterations, a decrypted image with resultant $CC = 0.9749$ is obtained as shown in Fig. 5(b), which means that most information about the plaintext has been extracted. Since only diffraction intensity maps are available and multiple phase-only masks are applied during the encryption, the proposed virtual-optics-based optical encoding strategy can effectively endure the attacks.^{21–24}

During image decryption, performance of security keys and robustness of the proposed method are further investigated. Figures 6(a) and 6(b) show decrypted images without the MIART and using wrong iteration number in the MIART, respectively. The maximum iteration number in the retrieval algorithm is preset as 1500, and an iteration error of 1 is set at all levels of the MIART for Fig. 6(b). Figure 6(c) shows a decrypted image after 1500 iterations using a wrong angle β during the decryption. The wrong angle β in the gyrator transform is set as 0.3π , and in this case other security keys are correct. The CC values are 0.0042, 0.0003, and -0.0003 for Figs. 6(a)–6(c), respectively. It can be seen in Figs. 6(a)–6(c) that when security keys are wrong or not properly applied, plaintext information cannot be extracted. For the sake of brevity, performance of other security keys, such as angle α , is not presented here. Since contamination might not be fully avoided during data storage or transmission, tolerance to ciphertext occlusion is also investigated. Figure 6(d) shows a decrypted image after 50 iterations, when 12.5% occlusions exist at each ciphertext. The CC value for Fig. 6(d) is 0.4632. It can be seen in Fig. 6(d) that plaintext information can still be observed, when ciphertexts are contaminated.

4 Conclusions

We have proposed a novel method for optical image encryption using multilevel ART and rotatable-phase-mask noninterferometric imaging. During the decryption, an iterative retrieval algorithm is further developed. The results demonstrate that the proposed method is feasible and effective for image encryption, and the proposed optical encryption scheme possesses high security and high robustness. The

strategy based on rotatable phase-only mask can provide a new and effective alternative for data/image encryption in the noninterferometric imaging, and it is highly expected that the proposed optical encryption method can also be applied to other case studies, such as color information encryption. In addition, watermarking technologies^{36–39} can be further applied as complementary methods for the transmission or storage of encrypted data/images.

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References

1. G. J. Simmons, "Symmetric and asymmetric encryption," *ACM Comput. Surv.* **11**, 305–330 (1979).
2. W. Diffie and M. Hellman, "New directions in cryptography," *IEEE Trans. Inf. Theory* **22**, 644–654 (1976).
3. P. Refregier and B. Javidi, "Optical image encryption based on input plane and Fourier plane random encoding," *Opt. Lett.* **20**, 767–769 (1995).
4. O. Matoba, T. Nomura, E. P. Cabré, M. S. Millán, and B. Javidi, "Optical techniques for information security," *Proc. IEEE* **97**, 1128–1148 (2009).
5. S. C. Pei and W. L. Hsue, "The multiple-parameter discrete fractional Fourier transform," *IEEE Signal Process. Lett.* **13**, 329–332 (2006).
6. S. C. Pei and W. L. Hsue, "Random discrete fractional Fourier transform," *IEEE Signal Process. Lett.* **16**, 1015–1018 (2009).
7. P. Clemente, V. Durán, V. T. Company, E. Tajahuerce, and J. Lancis, "Optical encryption based on computational ghost imaging," *Opt. Lett.* **35**, 2391–2393 (2010).
8. G. Situ and J. Zhang, "Double random-phase encoding in the Fresnel domain," *Opt. Lett.* **29**, 1584–1586 (2004).
9. G. Unnikrishnan, J. Joseph, and K. Singh, "Optical encryption by double-random phase encoding in the fractional Fourier domain," *Opt. Lett.* **25**, 887–889 (2000).
10. H. Kim, D. H. Kim, and Y. H. Lee, "Encryption of digital hologram of 3D object by virtual optics," *Opt. Express* **12**, 4912–4921 (2004).
11. X. F. Meng, L. Z. Cai, X. F. Xu, X. L. Yang, X. X. Shen, G. Y. Dong, and Y. R. Wang, "Two-step phase-shifting interferometry and its application in image encryption," *Opt. Lett.* **31**, 1414–1416 (2006).
12. H. T. Chang, W. C. Lu, and C. J. Kuo, "Multiple-phase retrieval for optical security systems by use of random-phase encoding," *Appl. Opt.* **41**, 4825–4834 (2002).
13. Z. Liu, L. Xu, C. Lin, and S. Liu, "Image encryption by encoding with a nonuniform optical beam in gyrator transform domains," *Appl. Opt.* **49**, 5632–5637 (2010).
14. H. E. Hwang, H. T. Chang, and W. N. Lie, "Multiple-image encryption and multiplexing using a modified Gerchberg-Saxton algorithm and phase modulation in Fresnel-transform domain," *Opt. Lett.* **34**, 3917–3919 (2009).
15. M. He, Q. Tan, L. Cao, Q. He, and G. Jin, "Security enhanced optical encryption system by random phase key and permutation key," *Opt. Express* **17**, 22462–22473 (2009).
16. T. J. Naughton, B. M. Hennelly, and T. Dowling, "Introducing secure modes of operation for optical encryption," *J. Opt. Soc. Am. A* **25**, 2608–2617 (2008).
17. H. Li and Y. Wang, "Double-image encryption based on iterative gyrator transform," *Opt. Commun.* **281**, 5745–5749 (2008).
18. Y. Zhang and B. Wang, "Optical image encryption based on interference," *Opt. Lett.* **33**, 2443–2445 (2008).
19. Y. Shi, G. Situ, and J. Zhang, "Multiple-image hiding in the Fresnel domain," *Opt. Lett.* **32**, 1914–1916 (2007).
20. Z. Liu, Q. Guo, L. Xu, M. A. Ahmad, and S. Liu, "Double image encryption by using iterative random binary encoding in gyrator domains," *Opt. Express* **18**, 12033–12043 (2010).
21. A. M. Youssef, "On the security of a cryptosystem based on multiple-parameters discrete fractional Fourier transform," *IEEE Signal Process. Lett.* **15**, 77–78 (2008).
22. W. Stallings, *Cryptography and Network Security: Principles and Practice*, 4th ed. Prentice Hall, New Jersey (2006).
23. X. Peng, P. Zhang, H. Wei, and B. Yu, "Known-plaintext attack on optical encryption based on double random phase keys," *Opt. Lett.* **31**, 1044–1046 (2006).
24. X. Peng, H. Wei, and P. Zhang, "Chosen-plaintext attack on lensless double-random phase encoding in the Fresnel domain," *Opt. Lett.* **31**, 3261–3263 (2006).

25. R. W. Gerchberg and W. O. Saxton, "A practical algorithm for the determination of phase from image and diffraction plane pictures," *Optik (Stuttgart)* **35**, 237–246 (1972).
26. J. R. Fienup, "Phase retrieval algorithms: a comparison," *Appl. Opt.* **21**, 2758–2769 (1982).
27. J. Miao, P. Charalambous, J. Kirz, and D. Sayre, "Extending the methodology of X-ray crystallography to allow imaging of micrometer-sized non-crystalline specimens," *Nature (London)* **400**, 342–344 (1999).
28. F. J. Dyson and H. Falk, "Period of a discrete cat mapping," *Am. Math. Mon.* **99**, 603–614 (1992).
29. W. Chen and X. Chen, "Space-based optical image encryption," *Opt. Express* **18**, 27095–27104 (2010).
30. W. Chen, C. Quan, and C. J. Tay, "Optical color image encryption based on Arnold transform and interference method," *Opt. Commun.* **282**, 3680–3685 (2009).
31. M. E. Testorf, B. M. Hennelly, and J. Ojeda-Castañeda, *Phase-Space Optics: Fundamentals and Applications*, McGraw-Hill, New York (2010).
32. W. Chen, X. Chen, and C. J. R. Sheppard, "Optical image encryption based on diffractive imaging," *Opt. Lett.* **35**, 3817–3819 (2010).
33. W. Chen and X. Chen, "Optical cryptography topology based on a three-dimensional particle-like distribution and diffractive imaging," *Opt. Express* **19**, 9008–9019 (2011).
34. C. J. Tay, C. Quan, W. Chen, and Y. Fu, "Color image encryption based on interference and virtual optics," *Opt. Laser Technol.* **42**, 409–415 (2010).
35. The plaintext (Lena): <http://sipi.usc.edu/database>.
36. X. Zhou and J. G. Chen, "Information hiding based on double-random phase encoding technology," *J. Mod. Opt.* **53**, 1777–1783 (2006).
37. X. F. Meng, L. Z. Cai, X. L. Yang, X. F. Xu, G. Y. Dong, X. X. Shen, H. Zhang, and Y. R. Wang, "Digital color image watermarking based on phase-shifting interferometry and neighboring pixel value subtraction algorithm in the discrete-cosine-transform domain," *Appl. Opt.* **46**, 4694–4701 (2007).
38. J. Li, "Robust image watermarking scheme against geometric attacks using a computer-generated hologram," *Appl. Opt.* **49**, 6302–6312 (2010).
39. X. Shi and D. Zhao, "Color image hiding based on the phase retrieval technique and Arnold transform," *Appl. Opt.* **50**, 2134–2139 (2011).



retrieval, optical image encryption, information security, diffractive imaging, optical microscopy, interferometric, and noninterferometric imaging.



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